

DEUTSCHE BANK INDIA

Retail Banking — Digital Payments

FUNCTIONAL REQUIREMENTS SPECIFICATION

UPI Enablement Platform

Field-Level Specification of Business Requirements Document (BRD v1.0)

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Document Control

1.1 Version History

Version	Date	Author	Description of Change
0.1	10-Jul-2026	Business Analyst	Initial FRS structure derived from BRD v1.0
1.0	date TBD	Business Analyst	Baseline version for technical design handoff

1.2 Approvals

Name	Role	Approval Status
[Solution Architect]	Technical Sign-off	Pending
[Head of Digital Banking]	Business Sign-off	Pending
[QA Lead]	Testability Sign-off	Pending

1.3 Relationship to Other Deliverables

This FRS elaborates the functional requirements (FR-xx) and business rules (BR-xx) defined in the Business Requirements Document (BRD v1.0) into implementation-ready, field-level specifications: screen and field definitions, validation rules, step-by-step processing logic, exception handling, and integration touchpoints. Where this FRS and the BRD conflict, the BRD's business intent governs and this FRS shall be corrected via change control. Use Cases, Wireframes, BPMN Process Flows, Sequence Diagrams, the ERD, and the API Integration Specification are companion deliverables that provide complementary detail (visual flow, data model, and API contract respectively) and should be read alongside this FRS.

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2. Purpose, Conventions, and Traceability

2.1 Purpose

This document specifies, for each functional requirement in scope, the precise screen/field behaviour, data validation, business logic, system response, and exception handling required to build and test the UPI Enablement Platform. It is written for API Developers, Mobile Developers, the Database Team, and the QA Team, and is the primary reference for sprint-level story elaboration and test-case design.

2.2 Specification Template

Each requirement in this FRS follows a consistent structure:

- Requirement ID and Name — mapped 1:1 to the corresponding BRD Functional Requirement ID
- Description — precise statement of expected system behaviour
- Actor(s) — who initiates or is affected by the requirement
- Pre-conditions — system/data state required before the requirement can execute
- Input Fields — field name, data type, mandatory/optional, format, validation rule
- Processing Logic — numbered step-by-step system behaviour
- Output / System Response — what is returned or displayed on success
- Exception Handling — error conditions, error codes, and system/customer-facing messages
- Post-conditions — resulting system/data state after successful execution
- Related Business Rules — cross-reference to BR-xx in the BRD

2.3 Field Validation Notation

Notation	Meaning
M	Mandatory field
O	Optional field
N(n)	Numeric field, maximum n digits
AN(n)	Alphanumeric field, maximum n characters
Enum{...}	Field restricted to the listed enumerated values
Regex	Field validated against a defined regular expression pattern

2.4 Error Code Convention

System error codes follow the pattern ERR-<MODULE>-<NNN> (e.g., ERR-REG-001). A consolidated Error Code Catalogue is provided in Section 8 (Appendix) at the end of this document. NPCI-originated error codes are passed through unmodified and mapped to a customer-facing message per the mapping table in the same appendix.

3. Module 1 — Customer Registration

Elaborates BRD Functional Requirements FR-REG-01 to FR-REG-12 and Business Rules BR-REG-01 to BR-REG-07.

3.1 Screen Field Specification — Mobile Verification

Field	Type / Validation	Mandatory	Notes
Registered Mobile Number	N(10), auto-fetched from CIF	M	Read-only; displayed masked (e.g., XXXXXX1234)
OTP	N(6), Regex <code>^[0-9]{6}\$</code>	M	6-digit numeric, expires in 300 seconds
Resend OTP	Action trigger	-	Enabled after 30-second cooldown; max 3 resends per session

3.2 Screen Field Specification — Consent Capture

Field	Type / Validation	Mandatory	Notes
Consent Text Version	AN(10), system-generated	M	Version ID logged against consent record (BR-REG-05)
Accept / Decline	Enum{Accept, Decline}	M	Decline halts registration flow (Alternate Flow 3a)

3.3 Screen Field Specification — Account Selection

Field	Type / Validation	Mandatory	Notes
Eligible Account List	System-generated list	M	Populated from CBS/CIF discovery; excludes ineligible types per BR-REG-06/07
Default Account Selector	Radio selection, single value	M	Exactly one account must be marked default

FR-REG-01 — Initiate UPI Registration

Description: The system shall present a 'Register for UPI' entry point on the mobile app home screen or UPI menu, visible only to customers not yet UPI-registered.

Actor: Retail Customer. Pre-conditions: Customer is logged into mobile banking with a valid session.

Processing Logic: 1. System checks customer's UPI registration status via UPI_Profile lookup. 2. If no active profile exists, display 'Register for UPI' CTA. 3. On tap, navigate to Mobile Verification screen (3.1).

Output: Navigation to Mobile Verification screen. Post-conditions: None (read-only check). Related Business Rules: BR-REG-01.

FR-REG-02 / FR-REG-03 — Mobile Number and Login Verification

Description: The system shall verify the customer's registered mobile number via OTP and confirm the customer is authenticated via existing mobile banking session or biometric re-auth.

Processing Logic: 1. System auto-fetches registered mobile number from CIF (masked display). 2. System triggers OTP dispatch via SMS Gateway. 3. Customer enters OTP. 4. System validates OTP against issued value and expiry (300s). 5. On success, proceed to Consent screen; on failure, display error and allow retry (max 5 attempts before 15-minute cooldown).

Exception Handling — Mobile Verification

Error Code	Condition	Customer-Facing Message
ERR-REG-001	OTP expired	This code has expired. Please request a new one.
ERR-REG-002	OTP mismatch	Incorrect code. Please try again.
ERR-REG-003	Max OTP attempts exceeded	Too many incorrect attempts. Please try again in 15 minutes.
ERR-REG-004	Mobile number mismatch with CBS/CIF record	We couldn't verify your number. Please visit a branch or contact support.

FR-REG-04 — Device Binding

Description: The system shall capture and bind device identifiers (device model, OS type, hashed device/IMEI identifier where permitted) to the UPI profile at registration.

Processing Logic: 1. Mobile app collects device metadata via SDK. 2. System hashes sensitive identifiers before transmission/storage (never stores raw IMEI). 3. System creates a Device record linked to the UPI_Profile with binding_status = Active. 4. If the customer later registers from a new device, the prior device binding is marked Inactive and the new device triggers re-verification (BR-REG-04).

Post-conditions: Device record created with binding_status = Active; audit log entry created.

FR-REG-05 / FR-REG-06 — Consent Capture and Logging

Description: The system shall present consent language (data sharing with NPCI/PSP) and log an immutable consent record.

Processing Logic: 1. Display current consent_version text. 2. Capture Accept/Decline. 3. On Accept, create Consent record with customer_id, consent_type = 'UPI_Registration', consent_version, timestamp, and channel = 'Mobile App'. 4. On Decline, halt flow and return to UPI landing screen with an explanatory message; no Consent record with Accept status is created.

FR-REG-07 / FR-REG-08 — Profile Fetch and Account Discovery

Description: The system shall fetch the customer profile from CIF and discover UPI-eligible accounts linked to the verified mobile number.

Processing Logic: 1. Call CIF API with customer_id to retrieve profile attributes. 2. Call CBS API to list accounts for customer_id. 3. Filter list to active savings/current accounts only (excludes loan, FD, credit card — BR-ACC accounts by type). 4. Return eligible account list to Account Selection screen. 5. If zero eligible accounts returned, display ERR-REG-005 and terminate registration flow.

Error Code	Condition	Customer-Facing Message
ERR-REG-005	No eligible accounts found	We couldn't find an eligible account. Please visit a branch or contact support.

Error Code	Condition	Customer-Facing Message
ERR-REG-006	CBS/CIF service timeout	We're unable to complete this right now. Please try again shortly.

FR-REG-09 / FR-REG-11 — Profile Creation and Success Confirmation

Description: The system shall create the UPI_Profile record, link it to the customer's CBS Customer ID, assign a VPA, and display a success confirmation.

Processing Logic: 1. System generates a default VPA in the format <mobile_or_alias>@dbsindia (subject to Bank naming convention and NPCI namespace rules). 2. System creates UPI_Profile record with profile_status = Active, linked to customer_id. 3. System creates Linked_Account record(s) for the selected account(s), with is_default flag set per customer selection. 4. System displays success screen with the assigned VPA and a 'Set PIN now' call to action. 5. System triggers a registration-confirmation notification (FR-NOT-05).

Post-conditions: UPI_Profile Active; Linked_Account record(s) created; audit log entry created; registration-confirmation notification queued.

FR-REG-10 — Custom UPI ID (Should Have)

Description: The system shall allow the customer to propose a custom VPA handle, subject to availability check against the NPCI/PSP namespace and Bank naming convention (alphanumeric, 3–50 characters, no profanity/reserved words).

Processing Logic: 1. Customer enters desired handle. 2. System checks availability via PSP switch. 3. If available, reserve and assign; if unavailable, display suggestions or prompt re-entry.

FR-REG-12 — Registration Rejection Handling

Description: The system shall reject registration where the mobile number does not match CBS/CIF records, with a clear remediation path (see ERR-REG-004 above).

4. Module 2 — Bank Account Linking

Elaborates BRD Functional Requirements FR-ACC-01 to FR-ACC-08 and Business Rules BR-ACC-01 to BR-ACC-05.

4.1 Screen Field Specification — Manage Linked Accounts

Field	Type / Validation	Mandatory	Notes
Linked Account List	System-generated	M	Displays masked account number, account nickname, default indicator
Add Account	Action trigger	-	Opens eligible-account discovery (re-uses FR-REG-07/08 logic)
Remove Account	Action trigger	-	Disabled for default account unless another account is set default first (BR-ACC-04)
Set as Default	Action trigger	-	Enabled for any non-default, active linked account

FR-ACC-01 / FR-ACC-08 — Fetch Eligible Accounts and Type Filtering

Description: The system shall fetch and display all eligible savings and current accounts, excluding ineligible account types.

Processing Logic: 1. Call CBS API to list all accounts for customer_id. 2. Filter to account_type IN {Savings, Current} AND account_status = Active. 3. Exclude Loan, Fixed Deposit, Credit Card, and Recurring Deposit account types (BR-ACC / ineligible type rule). 4. Return filtered list.

FR-ACC-02 / FR-ACC-05 — Default Account Designation and Change

Description: The system shall require exactly one linked account to be marked default, and shall allow the customer to change the default at any time.

Processing Logic: 1. On first linking, if only one account is selected, it is automatically set as default. 2. If multiple accounts selected, customer explicitly designates one as default. 3. On change-default action, system updates is_default = true on the newly selected Linked_Account record and is_default = false on the prior default, within a single atomic transaction.

FR-ACC-03 / FR-ACC-04 — Add / Remove Secondary Account

Description: The system shall allow adding secondary eligible accounts and removing any non-default linked account; removal of the default account requires re-assignment first.

Error Code	Condition	Customer-Facing Message
ERR-ACC-001	Attempt to remove default account with other accounts still linked	Please set another account as default before removing this one.
ERR-ACC-002	Attempt to link an ineligible account type	This account type isn't eligible for UPI.
ERR-ACC-003	Attempt to link an already-linked account	This account is already linked to your UPI profile.

FR-ACC-06 — Account Nickname Display

Description: The system shall display a customer-friendly nickname (or masked account number if no nickname is set) alongside each linked account throughout the UPI module.

FR-ACC-07 — Automatic Delinking

Description: The system shall detect accounts that become closed, frozen, or dormant (via CBS status change event or nightly reconciliation batch) and automatically delink them.

Processing Logic: 1. System receives or polls CBS account status. 2. On detecting status change to Closed/Frozen/Dormant, set Linked_Account.link_status = Delinked. 3. If the delinked account was default and other accounts remain linked, prompt customer to select a new default on next app open. 4. Trigger a security/account notification informing the customer of the delinking event.

Post-conditions: Linked_Account.link_status = Delinked; audit log entry created; notification queued.

5. Module 3 — UPI PIN Setup and Management

Elaborates BRD Functional Requirements FR-PIN-01 to FR-PIN-08 and Business Rules BR-PIN-01 to BR-PIN-06.

5.1 Screen Field Specification — PIN Setup

Field	Type / Validation	Mandatory	Notes
Debit Card Last 6 Digits	N(6)	M (or Aadhaar-OTP path)	Validated against CBS card record
Card Expiry (MM/YY)	N(4), Regex <code>^(0[1-9] 1[0-2])[0-9]{2}\$</code>	M (or Aadhaar-OTP path)	Must not be expired
OTP	N(6)	M	Sent to registered mobile; 300-second expiry
New PIN	N(4) or N(6) per scheme config	M	Masked entry; no sequential/repeated-digit PIN allowed (e.g., 1234, 1111)
Confirm PIN	N(4) or N(6)	M	Must match New PIN exactly

FR-PIN-01 / FR-PIN-02 — Debit Card and OTP Validation

Description: The system shall validate the customer's debit card (last 6 digits + expiry) against CBS card records, or accept Aadhaar-OTP as an NPCI-approved alternate method, and validate a mobile OTP, before allowing PIN creation.

Processing Logic: 1. Customer selects validation method (Debit Card or Aadhaar-OTP). 2. For Debit Card path: system calls CBS card-validation API with masked card details; on match, proceed. 3. System dispatches OTP via SMS Gateway. 4. Customer enters OTP; system validates against issued value and 300-second expiry. 5. On success, proceed to PIN entry.

Exception Handling — PIN Validation

Error Code	Condition	Customer-Facing Message
ERR-PIN-001	Debit card details do not match CBS record	The card details entered don't match our records.
ERR-PIN-002	Card expired	This card has expired. Please use an active card or Aadhaar verification.
ERR-PIN-003	OTP expired or mismatched	Incorrect or expired code. Please try again.

FR-PIN-03 / FR-PIN-04 — Secure PIN Entry and Encryption

Description: The system shall present a masked, tamper-resistant PIN pad, encrypt the PIN on-device prior to transmission, and never persist the PIN in plaintext at any layer (BR-PIN-01).

Processing Logic: 1. Customer enters New PIN and Confirm PIN via masked pad. 2. App validates New PIN = Confirm PIN client-side. 3. App validates PIN is not a trivially guessable pattern (sequential or repeated digits) client-side, with server-side re-validation. 4. App encrypts PIN block using the NPCI-mandated encryption scheme (device public key handshake). 5. Encrypted PIN block is transmitted to the PSP switch for NPCI-side PIN set, which returns only a

success/failure status — the Bank's own systems never receive or store the raw or reversible PIN, only pin_hash metadata for lockout tracking where applicable to the PSP model.

Post-conditions: UPI_PIN record created/updated with pin_set_date; failed_attempt_count reset to 0; audit log entry created (event only, no PIN value logged).

FR-PIN-05 — PIN Reset

Description: The system shall allow PIN reset via debit card/OTP validation without requiring the old PIN.

Processing Logic: Re-uses FR-PIN-01/02 validation flow, then proceeds directly to New PIN entry (FR-PIN-03/04) without an Old PIN step.

FR-PIN-06 — PIN Change

Description: The system shall allow PIN change by first validating the existing PIN.

Field	Type / Validation	Mandatory	Notes
Old PIN	N(4) or N(6)	M	Validated via PSP/NPCI PIN-verify call before allowing new PIN entry

FR-PIN-07 / FR-PIN-08 — Lockout and Recovery

Description: The system shall lock the UPI profile for 24 hours after 3 consecutive incorrect PIN entries (BR-PIN-03) and provide a clear unlock/reset path.

Processing Logic: 1. On each incorrect PIN entry during a transaction, increment UPI_PIN.failed_attempt_count. 2. On the 3rd consecutive failure, set lock_status = Locked, record lock timestamp. 3. Block all PIN-authenticated actions (payments, balance check, PIN change) while locked. 4. Auto-unlock after 24 hours, or allow immediate unlock via the PIN Reset flow (FR-PIN-05), which also resets failed_attempt_count to 0. 5. Trigger a security alert notification (FR-NOT-06) at the point of lockout.

Error Code	Condition	Customer-Facing Message
ERR-PIN-004	Incorrect PIN (1st or 2nd attempt)	Incorrect PIN. You have {n} attempt(s) remaining.
ERR-PIN-005	Profile locked after 3rd incorrect attempt	Your UPI PIN is locked for security. Reset your PIN or try again after 24 hours.
ERR-PIN-006	Action attempted while locked	Your UPI access is temporarily locked. Please reset your PIN to continue.

6. Module 4 — Send Money

Elaborates BRD Functional Requirements FR-TXN-01 to FR-TXN-14 and Business Rules BR-TXN-01 to BR-TXN-08.

6.1 Screen Field Specification — Select Payment Method

Field	Type / Validation	Mandatory	Notes
Payment Method Tab	Enum{Mobile, UPI ID, QR, Bank+IFSC, Self}	M	Determines subsequent field set
Mobile Number	N(10)	M (if Mobile tab)	Must resolve to a registered UPI mobile number via NPCI lookup
UPI ID (VPA)	AN(50), Regex <code>^[\w.-]{2,49}@[\\w]{2,49}\$</code>	M (if UPI ID tab)	Validated for format then resolved via NPCI/PSP
Account Number	N(6-17)	M (if Bank+IFSC tab)	Combined with IFSC for account-based transfer
IFSC Code	AN(11), Regex <code>^[A-Z]{4}0[A-Z0-9]{6}\$</code>	M (if Bank+IFSC tab)	Validated against IFSC master
Source Account (Self)	Dropdown of linked accounts	M (if Self tab)	Excludes the account already selected as destination

6.2 Screen Field Specification — Enter Amount & Authorise

Field	Type / Validation	Mandatory	Notes
Amount	Decimal(10,2), > 0	M	Validated against per-transaction limit (BR-TXN-01) before submission
Note / Remark	AN(50)	O	Free text, passed through to NPCI remarks field where supported
Recipient Name (display)	Read-only	-	Populated from NPCI VPA-resolution response for customer confirmation
UPI PIN	N(4) or N(6), masked	M	Required to authorise every outbound transaction (FR-TXN-07)

FR-TXN-01 to FR-TXN-05 — Payee Resolution by Method

Description: The system shall resolve a payee via mobile number, UPI ID, QR code, bank account + IFSC, or self-account selection, and display the resolved recipient name for customer confirmation before proceeding (FR-TXN-06).

Processing Logic (Mobile Number path): 1. Customer enters mobile number. 2. System calls PSP switch to resolve the number to a VPA via NPCI. 3. If multiple VPAs are linked to the number, present a selection list. 4. If no VPA is found, display ERR-TXN-001.

Processing Logic (UPI ID path): 1. Customer enters VPA. 2. System validates format client-side. 3. System calls PSP switch VPA-resolution API. 4. On success, display resolved account holder name; on failure, display ERR-TXN-002.

Processing Logic (Bank Account + IFSC path): 1. Customer enters account number and IFSC. 2. System validates IFSC format and existence against IFSC master. 3. System calls PSP switch to initiate an account-based UPI transfer (NPCI-supported account-to-account path). 4. Recipient name confirmation is displayed where returned by the beneficiary bank; if not returned, a clear warning is shown before authorisation.

Processing Logic (Self-Transfer path): 1. System pre-populates the customer's own linked accounts, excluding the source account from the destination list. 2. Standard amount/PIN flow applies; payee-specific limits (BR-TXN-04) are not applied, but per-transaction/daily limits (BR-TXN-01/02) still apply (BR-TXN-07).

Exception Handling — Payee Resolution

Error Code	Condition	Customer-Facing Message
ERR-TXN-001	Mobile number not registered on UPI	This number isn't registered for UPI payments.
ERR-TXN-002	UPI ID does not resolve	We couldn't find this UPI ID. Please check and try again.
ERR-TXN-003	Invalid IFSC code	This IFSC code is invalid.
ERR-TXN-004	Recipient account/VPA blacklisted by NPCI or internal fraud list	This payment cannot be completed. Please contact support.

FR-TXN-07 to FR-TXN-10 — Authorisation, Balance, Fraud, and Limit Checks

Description: The system shall require UPI PIN authorisation and perform balance, fraud/velocity, and limit validation prior to submitting any transaction to NPCI.

Processing Logic: 1. Customer enters amount and optional note, reviews recipient confirmation, and enters UPI PIN. 2. App encrypts PIN block on-device (FR-PIN-04 mechanism). 3. System validates: (a) Amount $\leq$ per-transaction limit (BR-TXN-01); (b) Cumulative 24-hour transaction value + this amount $\leq$ daily cap (BR-TXN-02); (c) If new registration/new payee, amount $\leq$ reduced 24-hour limit (BR-TXN-03/04). 4. System calls CBS for real-time balance validation (sufficient funds check). 5. System calls Fraud Monitoring System with transaction context; receives Allow / Step-Up / Decline decision within 1 second (NFR-PERF target). 6. On Allow, system submits the transaction request to NPCI via the PSP switch. 7. On Step-Up, system prompts additional verification (e.g., re-enter PIN, short delay) before resubmission. 8. On Decline, transaction is halted with ERR-TXN-008.

Error Code	Condition	Customer-Facing Message
ERR-TXN-005	Amount exceeds per-transaction limit	This amount exceeds the maximum allowed per transaction (₹1,00,000).
ERR-TXN-006	Daily cumulative limit exceeded	You've reached your daily UPI transaction limit.
ERR-TXN-007	Insufficient balance	Insufficient balance in the selected account.

Error Code	Condition	Customer-Facing Message
ERR-TXN-008	Fraud engine decline	This transaction could not be completed for security reasons. Please contact support.
ERR-TXN-009	New registration/payee reduced-limit breach	New payees/accounts are limited to ₹5,000 for the first 24 hours.

FR-TXN-11 / FR-TXN-14 — Status Display and Timeout Reconciliation

Description: The system shall display real-time transaction status on receiving the NPCI/CBS response, and shall automatically initiate reconciliation for transactions that time out without a definitive response (BR-TXN-08).

Processing Logic: 1. System awaits NPCI response within the configured timeout window (aligned to NPCI SLA, typically ≤ 30 seconds). 2. On receiving Success, update Transaction.txn_status = Success, display confirmation with NPCI RRN, trigger notification (FR-NOT-01). 3. On receiving Failure, update txn_status = Failed with reason, display failure message, trigger notification (FR-NOT-02). 4. On timeout (no response within window), set txn_status = Pending and enqueue a Deferred Status Check job that polls NPCI/CBS reconciliation status at defined intervals until resolved (per NPCI reconciliation cycle, typically within T+1 working day). 5. If Deferred Status Check confirms the debit occurred without a corresponding credit confirmation, trigger an automatic reversal to the customer's account and notify the customer (BR-TXN-08, NFR-AVL-04).

Post-conditions: Transaction record finalised with terminal status (Success/Failed/Reversed); Transaction_Leg and PSP_Transaction_Log records created; audit log entry created; notification queued.

FR-TXN-12 / FR-TXN-13 — Favourites and Notes (Could Have)

Description: The system may allow saving a payee as a Favourite (creates a Beneficiary record) and attaching a note/remark up to 50 characters to a payment.

7. Module 5 — Receive Money (Collect Requests)

Elaborates BRD Functional Requirements FR-COL-01 to FR-COL-08 and Business Rules BR-COL-01 to BR-COL-05.

7.1 Screen Field Specification — Create Collect Request

Field	Type / Validation	Mandatory	Notes
Payer VPA / Mobile Number	AN(50) or N(10)	M	Resolved the same way as Send Money payee resolution
Amount	Decimal(10,2), > 0	M	Subject to standard transaction limit rules (BR-COL-03)
Note	AN(50)	O	Displayed to payer with the request
Expiry	System-set, 24 hours from creation	-	Not customer-editable (BR-COL-01)

FR-COL-01 / FR-COL-02 — Create and Notify

Description: The system shall allow a registered customer to create a collect request against a payer VPA/mobile number, and shall notify the payer in real time.

Processing Logic: 1. Requester enters payer identifier, amount, optional note. 2. System resolves payer VPA (re-uses Send Money resolution logic). 3. System creates Collect_Request record with status = Pending, created_at = now, expiry_at = now + 24h. 4. System submits the collect request to NPCI via PSP switch for delivery to the payer's UPI app/bank. 5. System sends a push/SMS notification to the payer (FR-NOT-03).

FR-COL-03 / FR-COL-04 — Approve and Reject

Description: The system shall require UPI PIN authentication for the payer to approve a collect request, and shall allow rejection with an optional reason.

Processing Logic (Approve): 1. Payer opens the collect request from Notification Centre or in-app prompt. 2. System displays requester name, amount, note. 3. Payer authenticates with UPI PIN (same validation/limit/fraud checks as FR-TXN-07 to FR-TXN-10 apply). 4. On success, system processes the debit exactly as a standard Send Money transaction, and updates Collect_Request.status = Approved.

Processing Logic (Reject): 1. Payer selects Reject, optionally enters a reason code. 2. System updates Collect_Request.status = Rejected. 3. System notifies the requester in real time with the rejection (and reason, if provided) (BR-COL-05).

Error Code	Condition	Customer-Facing Message
ERR-COL-001	Collect request already expired	This request has expired and can no longer be actioned.
ERR-COL-002	Collect request already actioned (approved/rejected)	This request has already been actioned.

FR-COL-05 — Automatic Expiry

Description: The system shall automatically expire a collect request 24 hours after creation if not approved or rejected (BR-COL-01).

Processing Logic: A scheduled batch job runs at a minimum hourly frequency, identifies Collect_Request records where status = Pending and expiry_at ≤ now, sets status = Expired, and triggers a notification to the requester.

FR-COL-06 / FR-COL-07 — Reminders and Cancellation

Description: The system shall allow the requester to send up to 3 reminder notifications for a pending request (BR-COL-04) and to cancel a pending request before it is actioned.

Processing Logic (Reminder): 1. Requester taps Remind on a Pending request. 2. System checks Collect_Reminder count for this collect_id < 3. 3. If below limit, create a Collect_Reminder record and resend notification to payer; if at limit, disable the Remind action with an explanatory tooltip.

FR-COL-08 — Status Display

Description: The system shall display the current status (Pending / Approved / Rejected / Expired) for all collect requests raised or received by the customer, with real-time updates.

8. Module 6 — QR Payments

Elaborates BRD Functional Requirements FR-QR-01 to FR-QR-06 and Business Rules BR-QR-01 to BR-QR-04.

8.1 Screen Field Specification — Generate QR

Field	Type / Validation	Mandatory	Notes
QR Type	Enum{Static, Dynamic}	M	Static: amount editable by payer; Dynamic: amount fixed (BR-QR-01/02)
Amount (Dynamic only)	Decimal(10,2), > 0	M (if Dynamic)	Embedded in QR payload; not editable post-generation
Validity Window (Dynamic only)	Minutes, system default 15	M (if Dynamic)	QR becomes unscannable after expiry
Linked Account	Dropdown of linked accounts	M	Defaults to default account; customer may override

FR-QR-01 / FR-QR-02 — Static and Dynamic QR Generation

Description: The system shall generate a static QR (payer enters amount) linked to the customer's chosen account, and a dynamic QR with a fixed, non-editable amount and defined validity window.

Processing Logic: 1. Customer selects QR type and (for Dynamic) enters amount. 2. System constructs the UPI QR payload per NPCI QR specification (VPA, payee name, amount [Dynamic only], transaction reference, merchant/person indicator). 3. System creates a QR_Code record. 4. System renders the QR image for display/download/share.

FR-QR-03 / FR-QR-04 — Scan and Merchant Validation

Description: The system shall allow scanning a third-party or merchant QR code and shall validate merchant QR codes against the NPCI merchant registry before allowing payment.

Processing Logic: 1. Customer opens Scan & Pay; camera captures and decodes the QR payload. 2. System parses payee VPA, amount (if present), and merchant indicator. 3. If merchant-type QR, system calls the PSP switch to validate the merchant VPA against the NPCI merchant registry and retrieve MCC. 4. If valid, proceed to amount entry (if not embedded) and PIN authorisation, following the same processing logic as FR-TXN-07 to FR-TXN-11. 5. If the merchant is unverified or blacklisted, block the transaction (ERR-QR-002).

Error Code	Condition	Customer-Facing Message
ERR-QR-001	QR code unreadable or malformed	We couldn't read this QR code. Please try again or enter details manually.
ERR-QR-002	Merchant unverified/blacklisted	This merchant could not be verified. Payment cannot proceed.
ERR-QR-003	Dynamic QR expired	This QR code has expired. Please ask for a new one.

FR-QR-05 / FR-QR-06 — Save QR and Blacklist Enforcement

Description: The system may allow saving/downloading a generated QR as an image, and shall block payment to any QR-resolved VPA flagged as blacklisted or unverified.

9. Module 7 — Balance Check

Elaborates BRD Functional Requirements FR-BAL-01 to FR-BAL-03 and Business Rules BR-BAL-01 to BR-BAL-02.

9.1 Screen Field Specification

Field	Type / Validation	Mandatory	Notes
Account Selector	Dropdown of linked accounts	M	Displays account nickname (FR-BAL-02)
UPI PIN	N(4) or N(6), masked	M	Required for every balance enquiry, no caching (BR-BAL-01)

FR-BAL-01 / FR-BAL-03 — PIN-Authenticated Balance Enquiry with Daily Cap

Description: The system shall require UPI PIN authentication for every balance enquiry and shall enforce the NPCI-mandated daily cap (currently 15/day) per customer.

Processing Logic: 1. Customer selects account and enters PIN. 2. System checks the customer's balance-enquiry count for the current day (rolling calendar day) against the cap. 3. If within cap, system submits a balance-enquiry request via PSP switch to NPCI/CBS. 4. On response, display Available Balance and Current Balance. 5. If cap exceeded, display ERR-BAL-001 without submitting the enquiry.

Error Code	Condition	Customer-Facing Message
ERR-BAL-001	Daily balance-enquiry cap exceeded	You've reached today's limit for balance checks. Please try again tomorrow.

Post-conditions: No persistent balance value is cached client-side beyond the current session view, per BR-BAL-01.

10. Module 8 — Transaction History

Elaborates BRD Functional Requirements FR-HIS-01 to FR-HIS-05 and Business Rule BR-HIS-01.

10.1 Screen Field Specification — History List & Filters

Field	Type / Validation	Mandatory	Notes
Date Range Filter	Date From / Date To	O	Defaults to most recent 6 months; max range 10 years (BR-HIS-01)
Status Filter	Enum{All, Success, Failed, Pending, Reversed}	O	Defaults to All
Amount Range Filter	Decimal Min / Max	O	Both optional; validated Min ≤ Max
Beneficiary Filter	AN(50) search text	O	Matches against payee VPA or name
Transaction Type Filter	Enum{P2P, P2M, Self, Collect}	O	Defaults to All
Export to PDF	Action trigger	-	Applies currently active filters to the export

FR-HIS-01 / FR-HIS-02 — List Display and Filtering

Description: The system shall display a chronological list of all UPI transactions and allow filtering by date range, status, amount range, beneficiary, and transaction type, individually or in combination.

Processing Logic: 1. On screen load, query Transaction records for the customer's `upi_profile_id`, default view = last 6 months, sorted descending by `initiated_at`. 2. On filter application, construct a combined query across all active filter criteria (AND logic across filter types). 3. Return paginated results (default page size 20) with infinite scroll or pagination control. 4. Each row displays date, counterparty, amount, direction (sent/received), and status badge.

FR-HIS-03 — PDF Export

Description: The system shall allow the customer to export transaction history for a selected period as a PDF statement.

Processing Logic: 1. Customer selects date range and taps Export. 2. System generates a PDF containing Bank letterhead, customer name and masked account number, statement period, and a tabular transaction listing matching the active filters. 3. System offers Download/Share options. 4. Export generation for ranges exceeding 12 months is processed asynchronously with an in-app notification when ready, to avoid blocking the UI thread.

FR-HIS-04 — Transaction Detail Drill-Down

Description: The system shall display full transaction detail on drill-down, including UPI transaction reference number, NPCI RRN, timestamp, status, and payer/payee VPA.

Field Displayed	Source
Transaction ID (internal)	Transaction.transaction_id
NPCI RRN	Transaction.npci_rrn
Status	Transaction.txn_status
Initiated At / Completed At	Transaction.initiated_at / completed_at

Field Displayed	Source
Payer VPA / Payee VPA	Resolved from Transaction and counterparty records
Note / Remark	Transaction.note (if provided)
Raise Dispute CTA	Enabled only for Success/Reversed transactions within 180 days (BR-DIS-01)

FR-HIS-05 — Long-Term Retention and Search

Description: The system shall retain and make searchable transaction history for a minimum of 10 years, with the default in-app view limited to the most recent 6 months for performance, and older records retrievable via explicit date-range search (BR-HIS-01).

Processing Logic: Historical records beyond an operational threshold (e.g., 12 months) may be moved to a lower-cost archive tier at the database layer, retrievable via the same search API with an expected higher (but bounded, per NFR-PERF-04 target) response time for archive queries.

11. Module 9 — Notifications

Elaborates BRD Functional Requirements FR-NOT-01 to FR-NOT-07 and Business Rules BR-NOT-01 to BR-NOT-02.

11.1 Notification Event Catalogue

Event Type	Trigger	Channel(s)	Opt-Out Permitted?
Payment Success (Sent)	Transaction.txn_status → Success (debit)	Push + SMS	No (BR-NOT-02)
Payment Success (Received)	Transaction.txn_status → Success (credit)	Push + SMS	No
Payment Failed	Transaction.txn_status → Failed	Push + SMS	No
Collect Request Received	New Collect_Request created for this customer as payer	Push + SMS	No
Collect Request Approved/Rejected/Expired	Collect_Request.status change	Push	No
Refund Processed	Refund record created/completed	Push + SMS	No
Chargeback Processed	Refund record linked to a dispute completed	Push + SMS	No
UPI Registration Confirmation	UPI_Profile created	Push + SMS	No
PIN Set / Changed / Reset	UPI_PIN event	Push + SMS	No
Security Alert (new device, PIN lockout)	Device binding change or lockout event	Push + SMS	No
Cross-Sell / Marketing Offer	Post-transaction contextual offer engine trigger	Push	Yes (FR-NOT-07)

FR-NOT-01 to FR-NOT-06 — Mandatory Event Notifications

Description: The system shall send a real-time push notification and SMS (per the catalogue above) for every payment, collect, refund/chargeback, registration/PIN, and security event, within 5 seconds of status determination (BR-NOT-01).

Processing Logic: 1. Triggering event occurs (e.g., Transaction status update). 2. Event is published to the Notification Service via an internal event queue. 3. Notification Service composes the message from a versioned template per event type. 4. Notification Service dispatches via Push provider and/or SMS Gateway per the channel matrix. 5. Delivery status is logged on the Notification record. 6. Failed push delivery falls back to SMS for security-critical events (PIN lockout, new device).

FR-NOT-07 — Notification Preferences

Description: The system shall allow the customer to opt in/out of non-mandatory (marketing/cross-sell) notifications via app settings; security and transaction-confirmation notifications remain non-optional (BR-NOT-02).

11.2 Screen Field Specification — Notification Preferences

Field	Type / Validation	Mandatory	Notes
Marketing/Cross-Sell Toggle	Boolean, default = On	-	Customer-editable
Transaction Notifications	Boolean, always On, read-only	-	Cannot be disabled (BR-NOT-02)
Security Alerts	Boolean, always On, read-only	-	Cannot be disabled

12. Module 10 — Dispute Management

Elaborates BRD Functional Requirements FR-DIS-01 to FR-DIS-08 and Business Rules BR-DIS-01 to BR-DIS-04.

12.1 Screen Field Specification — Raise Dispute

Field	Type / Validation	Mandatory	Notes
Transaction Reference	Auto-populated from Transaction Detail	M	Read-only; source transaction context
Dispute Category	Enum{Money debited, not credited to beneficiary; Goods/services not received; Duplicate transaction; Incorrect amount; Other}	M	Drives internal routing/priority
Description	AN(500)	M	Free-text customer explanation
Supporting Note/Attachment	File upload (image/PDF, max 5MB)	O	Optional evidence

FR-DIS-01 / FR-DIS-02 — Raise Dispute with Category

Description: The system shall allow a customer to raise a dispute against a completed transaction within 180 days of the transaction date, requiring selection of a dispute category (BR-DIS-01).

Processing Logic: 1. Customer taps 'Raise Dispute' from Transaction Detail (FR-HIS-04). 2. System validates transaction date is within 180 days; if not, disable the action (ERR-DIS-001 shown if attempted via deep link). 3. Customer selects category, enters description, optionally attaches supporting evidence. 4. On submit, system creates a Dispute record with status = Submitted.

FR-DIS-03 / FR-DIS-04 — Reference Number Generation

Description: The system shall generate a unique internal complaint reference number immediately upon submission, and shall capture the NPCI complaint/reference number once the dispute is escalated to NPCI.

Processing Logic: 1. On Dispute creation, system generates internal_ref_no in the format DBI-DIS-<YYYYMMDD>-<sequence>. 2. System routes the dispute to the Operations queue and, for categories requiring inter-bank resolution (e.g., money debited but not credited), submits the dispute to NPCI's dispute management system via the PSP switch. 3. On NPCI acknowledgement, system stores the returned npc_ref_no against the Dispute record and displays it to the customer.

FR-DIS-05 / FR-DIS-06 — Status Tracking and Refund Status

Description: The system shall allow the customer to track dispute status in real time and shall display refund status and expected/actual credit date where applicable.

Status Value	Meaning
Submitted	Dispute created, pending initial triage
Under Investigation	Assigned to Operations/NPCI for review
Escalated	Referred to NPCI or beneficiary bank for resolution beyond initial TAT
Resolved — Refunded	Dispute upheld; Refund record created and credited

Status Value	Meaning
Resolved — Rejected	Dispute investigated and not upheld, with reason provided

Processing Logic: 1. Dispute.status transitions are driven by Operations/Compliance actions (Dispute_Note entries) and/or NPCI status callbacks. 2. Each status change triggers a customer notification. 3. Where a Refund record is created and linked to the dispute, the Refund.refund_status and processed_at are surfaced on the tracking screen.

FR-DIS-07 — Automated TAT Escalation

Description: The system should provide an automated escalation workflow if a dispute is not resolved within the NPCI-mandated TAT (BR-DIS-03).

Processing Logic: A scheduled job identifies Dispute records where status NOT IN (Resolved-Refunded, Resolved-Rejected) and (now - raised_at) exceeds the category-specific TAT threshold; such disputes are flagged for supervisor escalation and, where applicable, auto-submitted to NPCI escalation channels.

FR-DIS-08 — Internal Investigation Notes

Description: The system shall allow Tier 2 Operations users to update dispute status and add investigation notes visible to Compliance and Audit (but not directly to the customer, unless explicitly shared as a customer-facing status comment).

Error Code	Condition	Customer-Facing Message
ERR-DIS-001	Dispute attempted outside 180-day window	This transaction is outside the eligible window for raising a dispute.
ERR-DIS-002	Duplicate dispute already open for this transaction	A dispute is already open for this transaction. You can track its status in Dispute Tracking.

13. Appendix — Consolidated Error Code Catalogue

This appendix consolidates all module-level error codes defined in Sections 3–12 for reference by developers and the QA team. Codes are grouped by module prefix.

Prefix	Module	Code Range
ERR-REG	Customer Registration	001–006
ERR-ACC	Bank Account Linking	001–003
ERR-PIN	UPI PIN Management	001–006
ERR-TXN	Send Money	001–009
ERR-COL	Collect Requests	001–002
ERR-QR	QR Payments	001–003
ERR-BAL	Balance Check	001
ERR-DIS	Dispute Management	001–002

13.1 NPCI Error Pass-Through

NPCI-originated error/response codes (e.g., U16, U17, ZM, Z9 and similar NPCI-defined codes) are received via the PSP switch and mapped to the nearest equivalent customer-facing message from the catalogue above where a direct mapping exists; where no direct mapping exists, the system shall display a generic fallback: 'This transaction could not be completed. Please try again or contact support.' The full NPCI-to-internal error mapping table is maintained by the Integration Team as a living reference and is out of scope for this business-facing FRS.

13.2 Traceability Note

Every functional requirement in this FRS (FR-xx) traces to a corresponding BRD Functional Requirement of the same ID, and every error/exception condition traces to a UAT scenario in the companion UAT Scenarios deliverable, per the Requirement Traceability Matrix (RTM).